

ETF Momentum Research

A reproducible, long-only test of 12-1 ETF momentum

Does volatility scaling and an SPY 200-day regime filter improve the risk-adjusted performance of monthly ETF momentum after realistic transaction costs?

24 liquid ETFs | 2011-2018 research | 2019-31 Aug 2026 out-of-sample | 10 bps one-way costs

Research design

A fixed design stated before reading the out-of-sample comparison

Universe

24 US-listed ETFs across equities, rates, credit, commodities, FX and sectors.

Signal

Prior 252-to-21 trading-day return. The most recent month is excluded.

Execution

Monthly rebalance; signal becomes active on the following trading day; 10 bps × one-way turnover.

- Benchmark: equal-weight available ETFs.
- Active variants: top-six equal weight, inverse volatility, and inverse volatility with SPY filter.
- Filter holds cash below the SPY 200-day moving average.

Bias controls and what remains

A backtest is evidence only if its assumptions are made visible

- Research period and out-of-sample period are separated.
- Weights are shifted by one trading day to avoid same-day signal execution.
- Turnover is recorded; costs are charged only when the portfolio changes.
- Raw prices are cached, weights and daily wealth paths are written to CSV.
- Remaining limitations: selected ETF universe, adjusted-close vendor conventions, stylised execution, no tax, impact or capacity model.

Research-period results

2011–2018; annualised values after 10 bps one-way transaction costs

Strategy	CAGR	Vol	Sharpe	Max DD	Turnover
Equal weight	5.4%	9.9%	0.58	−14.8%	0.1x
12–1 momentum	6.2%	11.5%	0.58	−23.8%	4.6x
Momentum inverse vol	6.6%	10.9%	0.64	−19.5%	5.7x
Momentum regime filter	7.0%	10.2%	0.71	−18.2%	5.5x

In the research period, the regime-filtered variant had the highest reported Sharpe. This is a hypothesis for testing—not a conclusion to market.

Out-of-sample results

2019–31 August 2026; the result changed the narrative

Strategy	CAGR	Vol	Sharpe	Max DD	Turnover
Equal weight	12.9%	12.8%	1.01	−26.9%	0.0x
12–1 momentum	17.6%	17.2%	1.03	−24.5%	5.1x
Momentum inverse vol	15.5%	15.0%	1.04	−23.2%	7.2x
Momentum regime filter	8.3%	12.2%	0.71	−17.9%	6.9x

Inverse-volatility momentum produced the highest out-of-sample Sharpe in this fixed run. The filter reduced volatility and drawdown, but also lowered CAGR.

What the evidence supports

The conclusion is narrower than the question

Supported

- The tested inverse-volatility implementation improved out-of-sample Sharpe versus the other tested variants.
- The regime filter traded return for lower realised risk in this sample.

Not supported

- A claim that the regime filter is generally superior.
- A claim of a deployable alpha after market impact, capacity or alternative-universe tests.

Next tests before stronger claims

The research agenda matters more than pretending the first model is finished

- Vary lookback, skip-month, number of holdings and rebalance frequency without tuning on the out-of-sample window.
- Test point-in-time or alternative ETF universes; inspect delisted / unavailable instruments where possible.
- Add cash yield, impact and liquidity proxies; test sensitivity to cost assumptions.
- Decompose returns by asset class, crisis periods and rate regimes.
- Use walk-forward evaluation and record every model choice.

Reproducibility and interview discussion

A portfolio project should be inspectable, not just persuasive

Repository deliverables

Cached raw data • source code • unit tests • weights • metrics • research report

- Explain the timing convention and the five largest sources of bias.
- Defend the choice of ETF universe and cost assumption.
- Describe why the out-of-sample result changed the narrative.
- State the next test that could disconfirm the idea.